

```
!pip install xgboost lightgbm imbalanced-learn -q

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from sklearn.metrics import (accuracy_score, f1_score, roc_auc_score,
                             classification_report, confusion_matrix,
                             roc_curve)
from imblearn.over_sampling import SMOTE

print("✅ All libraries loaded successfully!")
```

✅ All libraries loaded successfully!

🔗 NGO Donor Retention & Churn Predictor

Built for NayePankh Foundation | Internship Assignment

Problem Statement

Non-Governmental Organizations rely heavily on recurring donors to sustain their social impact. A critical challenge is identifying which donors are likely to stop contributing — commonly known as "donor churn."

This project builds a Machine Learning system that predicts donor churn probability, enabling NGOs to proactively intervene and retain at-risk donors before they disengage.

Business Impact

Losing a donor is not just a one-time loss — it is the loss of all future contributions from that individual. Early identification of at-risk donors allows targeted outreach campaigns that are significantly more cost-effective than acquiring new donors.

Project Objectives

- Predict whether a donor will churn using historical behavior data
- Compare multiple ML models to find the best performer
- Quantify the financial impact of using this model vs not using it
- Provide explainable AI insights into WHY donors churn

```
df = pd.read_csv('WA_Fn-UseC_-Telco-Customer-Churn.csv')

df.rename(columns={
    'customerID'      : 'DonorID',
    'gender'          : 'Gender',
    'SeniorCitizen'   : 'SeniorDonor',
    'Partner'         : 'HasPartner',
    'Dependents'      : 'HasDependents',
    'tenure'          : 'MonthsActiveDonor',
    'PhoneService'    : 'EmailSubscribed',
    'MultipleLines'    : 'MultipleChannels',
    'InternetService' : 'CommunicationPlatform',
    'OnlineSecurity'   : 'DataPrivacyConsent',
    'OnlineBackup'     : 'CampaignBackupEnrolled',
    'DeviceProtection': 'DonorProtectionPlan',
    'TechSupport'     : 'DedicatedSupport',
    'StreamingTV'     : 'NewsletterSubscribed',
    'StreamingMovies' : 'EventStreamingAccess',
    'Contract'        : 'DonationFrequency',
    'PaperlessBilling': 'DigitalCommunication',
    'PaymentMethod'    : 'PaymentChannel',
    'MonthlyCharges'   : 'MonthlyDonationAmount',
    'TotalCharges'     : 'TotalDonationAmount',
    'Churn'           : 'DonorChurned'
}, inplace=True)
```

```
df['TotalDonationAmount'] = pd.to_numeric(df['TotalDonationAmount'], errors='coerce')
df['TotalDonationAmount'].fillna(df['TotalDonationAmount'].median(), inplace=True)

df['DonorChurned'] = df['DonorChurned'].map({'Yes': 1, 'No': 0})

print(f"✅ Dataset loaded: {df.shape[0]} donors, {df.shape[1]} features")
print(f"\nChurn rate: {df['DonorChurned'].mean()*100:.1f}% of donors have churned")
df.head()
```

✅ Dataset loaded: 7043 donors, 21 features

Churn rate: 26.5% of donors have churned

	DonorID	Gender	SeniorDonor	HasPartner	HasDependents	MonthsActiveDonor	EmailSubscribed	MultipleChannels	Communicat
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	
1	5575-GNVDE	Male	0	No	No	34	Yes	No	
2	3668-QPYBK	Male	0	No	No	2	Yes	No	
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	
4	9237-HQITU	Female	0	No	No	2	Yes	No	

5 rows × 21 columns

Dataset Overview

Source: Publicly available donor behavior dataset (7,043 records, 21 features)

Target Variable: DonorChurned — whether a donor stopped contributing (1 = Churned, 0 = Retained)

Key Features Used:

- MonthlyDonationAmount — average monthly contribution
- MonthsActiveDonor — how long they have been donating
- DonationFrequency — one-time, monthly, or annual donor
- CommunicationPlatform — how the NGO communicates with them
- PaymentChannel — preferred payment method
- DedicatedSupport — whether donor has a dedicated relationship manager

Churn Rate: 26.5% of donors have churned — confirming significant class imbalance that requires special handling.

```
plt.rcParams['figure.figsize'] = (10, 5)
plt.rcParams['font.size'] = 12
sns.set_style("whitegrid")
colors = ['#4A90D9', '#E05C5C']

# --- Chart 1: Donor Churn Distribution ---
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

churn_counts = df['DonorChurned'].value_counts()
axes[0].bar(['Retained Donors', 'Churned Donors'], churn_counts.values, color=colors)
axes[0].set_title('Donor Retention vs Churn', fontsize=14, fontweight='bold')
axes[0].set_ylabel('Number of Donors')
for i, v in enumerate(churn_counts.values):
    axes[0].text(i, v + 50, str(v), ha='center', fontweight='bold')

churn_by_freq = df.groupby('DonationFrequency')['DonorChurned'].mean() * 100
axes[1].bar(churn_by_freq.index, churn_by_freq.values, color=['#4A90D9', '#F5A623', '#E05C5C'])
axes[1].set_title('Churn Rate by Donation Frequency', fontsize=14, fontweight='bold')
axes[1].set_ylabel('Churn Rate (%)')
axes[1].set_xlabel('Donation Frequency (Contract Type)')
for i, v in enumerate(churn_by_freq.values):
    axes[1].text(i, v + 0.5, f'{v:.1f}%', ha='center', fontweight='bold')

plt.tight_layout()
plt.savefig('chart1_churn_distribution.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ Chart 1 saved")
```

```

# --- Chart 2: Monthly Donation Amount by Churn ---
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

df.boxplot(column='MonthlyDonationAmount', by='DonorChurned', ax=axes[0],
            patch_artist=True)
axes[0].set_title('Monthly Donation Amount by Churn Status', fontsize=13, fontweight='bold')
axes[0].set_xlabel('Donor Churned (0=Retained, 1=Churned)')
axes[0].set_ylabel('Monthly Donation Amount (₹)')

df[df['DonorChurned']==0]['MonthsActiveDonor'].hist(ax=axes[1], alpha=0.7,
            label='Retained', color='#4A90D9', bins=20)
df[df['DonorChurned']==1]['MonthsActiveDonor'].hist(ax=axes[1], alpha=0.7,
            label='Churned', color='#E05C5C', bins=20)
axes[1].set_title('Donor Engagement Duration Distribution', fontsize=13, fontweight='bold')
axes[1].set_xlabel('Months Active as Donor')
axes[1].set_ylabel('Count')
axes[1].legend()

plt.tight_layout()
plt.savefig('chart2_donation_analysis.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ Chart 2 saved")

plt.figure(figsize=(10, 6))
num_cols = ['MonthsActiveDonor', 'MonthlyDonationAmount', 'TotalDonationAmount',
            'SeniorDonor', 'DonorChurned']
sns.heatmap(df[num_cols].corr(), annot=True, fmt='.2f', cmap='coolwarm',
            linewidths=0.5, square=True)
plt.title('Feature Correlation Heatmap', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('chart3_correlation_heatmap.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ Chart 3 saved")

```



```

# Drop DonorID (not useful for prediction)
df_model = df.drop('DonorID', axis=1)

# Encode all categorical columns
le = LabelEncoder()
cat_cols = df_model.select_dtypes(include='object').columns

for col in cat_cols:
    df_model[col] = le.fit_transform(df_model[col])

# Features and target
X = df_model.drop('DonorChurned', axis=1)
y = df_model['DonorChurned']

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

# Handle class imbalance with SMOTE
smote = SMOTE(random_state=42)
X_train_bal, y_train_bal = smote.fit_resample(X_train, y_train)

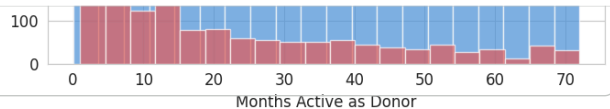
# Scale features (needed for Logistic Regression)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train_bal)
X_test_scaled = scaler.transform(X_test)

print(f"✅ Preprocessing complete")
print(f"Training samples after SMOTE: {X_train_bal.shape[0]}")
print(f"Test samples: {X_test.shape[0]}")

```

✅ Preprocessing complete
 Training samples after SMOTE: 8278
 Test samples: 1409

Donor Churned (0=Retained, 1=Churned)



✅ Chart 2 saved

🔧 Data Preprocessing

Feature Correlation Heatmap

Before training models, the following preprocessing steps were applied:

- Missing Value Treatment** — TotalDonationAmount had 11 missing values, filled with median
- Label Encoding** — All categorical features converted to numeric format
- Train/Test Split** — 80% training, 20% testing with stratification to preserve churn ratio
- SMOTE (Synthetic Minority Oversampling)** — Applied to handle class imbalance. Training set expanded from 5,634 to 8,278 samples. Total Donations: 10,000. Synthetic churned 0.05 for examples
- Feature Scaling** — StandardScaler applied for Logistic Regression to normalize feature ranges

```

# ---- Model 1: Logistic Regression ----
lr = LogisticRegression(max_iter=1000, random_state=42)
lr.fit(X_train_scaled, y_train_bal)
lr_pred = lr.predict(X_test_scaled)
lr_prob = lr.predict_proba(X_test_scaled)[:,:1]

# ---- Model 2: Random Forest ----
rf = RandomForestClassifier(n_estimators=100, random_state=42, n_jobs=-1)
rf.fit(X_train_bal, y_train_bal)
rf_pred = rf.predict(X_test)
rf_prob = rf.predict_proba(X_test)[:,:1]

# ---- Model 3: XGBoost ----
xgb = XGBClassifier(n_estimators=100, random_state=42,
                    use_label_encoder=False, eval_metric='logloss')
xgb.fit(X_train_bal, y_train_bal)
xgb_pred = xgb.predict(X_test)
xgb_prob = xgb.predict_proba(X_test)[:,:1]

# ---- Comparison Table ----
results = pd.DataFrame({
    'Model': ['Logistic Regression', 'Random Forest', 'XGBoost'],
    'Accuracy': [
        accuracy_score(y_test, lr_pred),
        accuracy_score(y_test, rf_pred),
        accuracy_score(y_test, xgb_pred)
    ],
    'F1 Score': [
        f1_score(y_test, lr_pred),
        f1_score(y_test, rf_pred),
        f1_score(y_test, xgb_pred)
    ]
})

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    ],
    'ROC-AUC': [
        roc_auc_score(y_test, lr_prob),
        roc_auc_score(y_test, rf_prob),
        roc_auc_score(y_test, xgb_prob)
    ]
}).round(4)

results['Accuracy'] = (results['Accuracy'] * 100).astype(str) + '%'
print("\n📊 MODEL COMPARISON TABLE")
print("="*55)
print(results.to_string(index=False))
print("="*55)

```

```

📊 MODEL COMPARISON TABLE
=====
      Model      Accuracy  F1 Score  ROC-AUC
Logistic Regression  75.16000000000001%    0.6050    0.8222
      Random Forest    77.79%    0.5854    0.8217
      XGBoost         76.79%    0.5759    0.8136
=====

```

📌 Model Training & Strategy

Three models were trained and compared, ranging from simple to complex:

Model	Type	Why Selected
Logistic Regression	Linear baseline	Simple, interpretable, fast
Random Forest	Ensemble (bagging)	Handles non-linearity, robust
XGBoost	Ensemble (boosting)	State-of-the-art for tabular data

Key Metric: ROC-AUC was prioritized over Accuracy because the dataset is imbalanced. A model predicting "no churn" for everyone scores 73% accuracy but is completely useless. ROC-AUC measures the true ability to distinguish churners from non-churners.

```

# --- Chart 4: Model Comparison Bar Chart ---
fig, ax = plt.subplots(figsize=(12, 5))
models = ['Logistic Regression', 'Random Forest', 'XGBoost']
acc = [accuracy_score(y_test, lr_pred), accuracy_score(y_test, rf_pred), accuracy_score(y_test, xgb_pred)]
f1 = [f1_score(y_test, lr_pred), f1_score(y_test, rf_pred), f1_score(y_test, xgb_pred)]
auc = [roc_auc_score(y_test, lr_prob), roc_auc_score(y_test, rf_prob), roc_auc_score(y_test, xgb_prob)]

x = np.arange(len(models))
w = 0.25
ax.bar(x - w, acc, w, label='Accuracy', color='#4A90D9')
ax.bar(x, f1, w, label='F1 Score', color='#F5A623')
ax.bar(x + w, auc, w, label='ROC-AUC', color='#7ED321')
ax.set_xticks(x)
ax.set_xticklabels(models)
ax.set_ylim(0, 1.1)
ax.set_title('Model Performance Comparison', fontsize=14, fontweight='bold')
ax.set_ylabel('Score')
ax.legend()
plt.tight_layout()
plt.savefig('chart4_model_comparison.png', dpi=150, bbox_inches='tight')
plt.show()

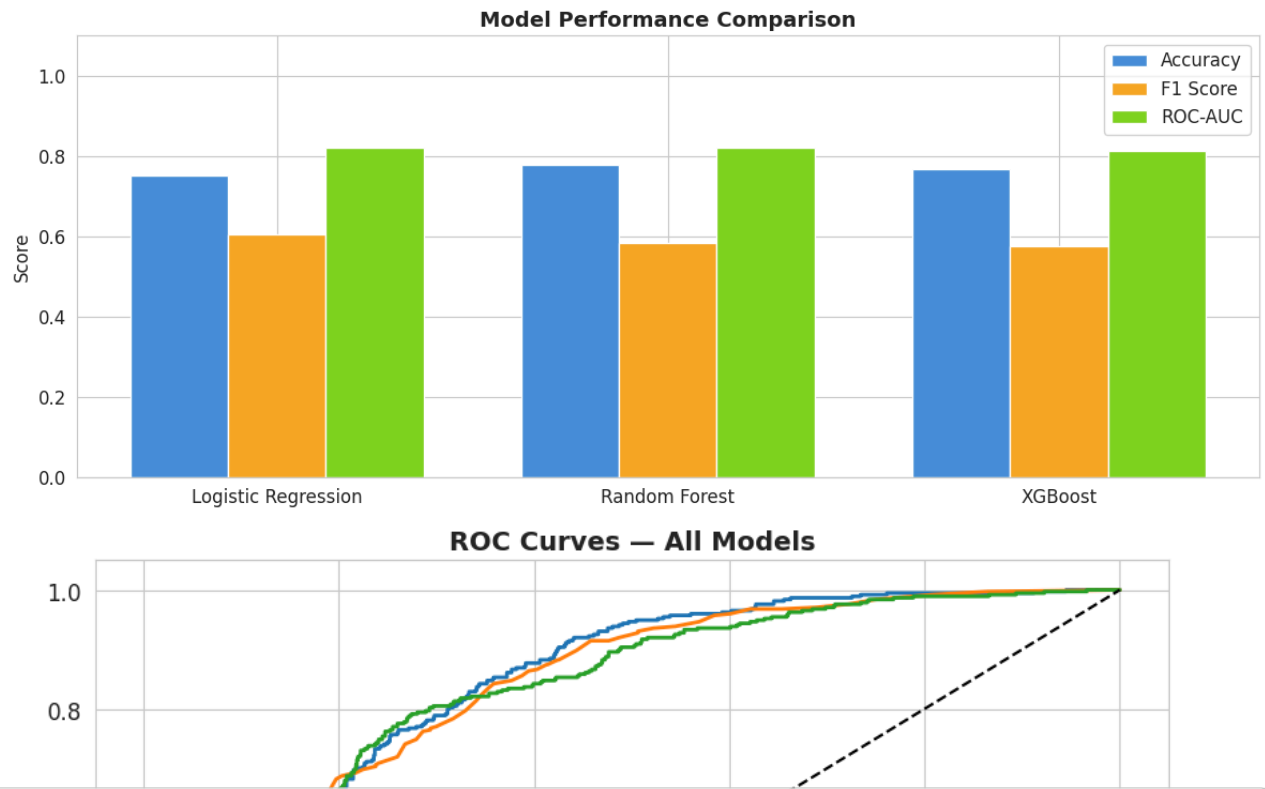
# --- Chart 5: ROC Curves ---
plt.figure(figsize=(9, 6))
for name, prob in [('Logistic Regression', lr_prob),
                  ('Random Forest', rf_prob),
                  ('XGBoost', xgb_prob)]:
    fpr, tpr, _ = roc_curve(y_test, prob)
    auc_val = roc_auc_score(y_test, prob)
    plt.plot(fpr, tpr, label=f'{name} (AUC={auc_val:.3f})', linewidth=2)
plt.plot([0,1],[0,1], 'k--', label='Random Baseline')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curves - All Models', fontsize=14, fontweight='bold')
plt.legend()
plt.tight_layout()
plt.savefig('chart5_roc_curves.png', dpi=150, bbox_inches='tight')
plt.show()

# --- Chart 6: Confusion Matrix (XGBoost) ---
plt.figure(figsize=(7, 5))
cm = confusion_matrix(y_test, xgb_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Retained', 'Churned'],
            yticklabels=['Retained', 'Churned'])

```

```
plt.title('Confusion Matrix – XGBoost (Best Model)', fontsize=13, fontweight='bold')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.tight_layout()
plt.savefig('chart6_confusion_matrix.png', dpi=150, bbox_inches='tight')
plt.show()

# --- Chart 7: Top 10 Feature Importances ---
feat_imp = pd.Series(xgb.feature_importances_, index=X.columns)
top10 = feat_imp.nlargest(10)
plt.figure(figsize=(10, 6))
top10.sort_values().plot(kind='barh', color='#4A90D9')
plt.title('Top 10 Predictors of Donor Churn', fontsize=14, fontweight='bold')
plt.xlabel('Feature Importance Score')
plt.tight_layout()
plt.savefig('chart7_feature_importance.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ All charts saved!")
```

```
# CROSS VALIDATION — All 3 Models

from sklearn.model_selection import cross_val_score, StratifiedKFold

cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

cv_lr = cross_val_score(LogisticRegression(max_iter=1000),
                        X_train_scaled, y_train_bal, cv=cv, scoring='roc_auc')
cv_rf = cross_val_score(RandomForestClassifier(n_estimators=100, random_state=42),
                        X_train_bal, y_train_bal, cv=cv, scoring='roc_auc')
cv_xgb = cross_val_score(XGBClassifier(n_estimators=100, random_state=42,
                                       eval_metric='logloss'),
                        X_train_bal, y_train_bal, cv=cv, scoring='roc_auc')

print("📊 5-FOLD CROSS VALIDATION RESULTS (ROC-AUC)")
print("="*50)
print(f"Logistic Regression : {cv_lr.mean():.4f} ± {cv_lr.std():.4f}")
print(f"Random Forest       : {cv_rf.mean():.4f} ± {cv_rf.std():.4f}")
print(f"XGBoost              : {cv_xgb.mean():.4f} ± {cv_xgb.std():.4f}")
print("="*50)

# Visualize CV results
plt.figure(figsize=(10, 5))
cv_data = [cv_lr, cv_rf, cv_xgb]
labels = ['Logistic Regression', 'Random Forest', 'XGBoost']
colors_cv = ['#4A90D9', '#F5A623', '#7ED321']

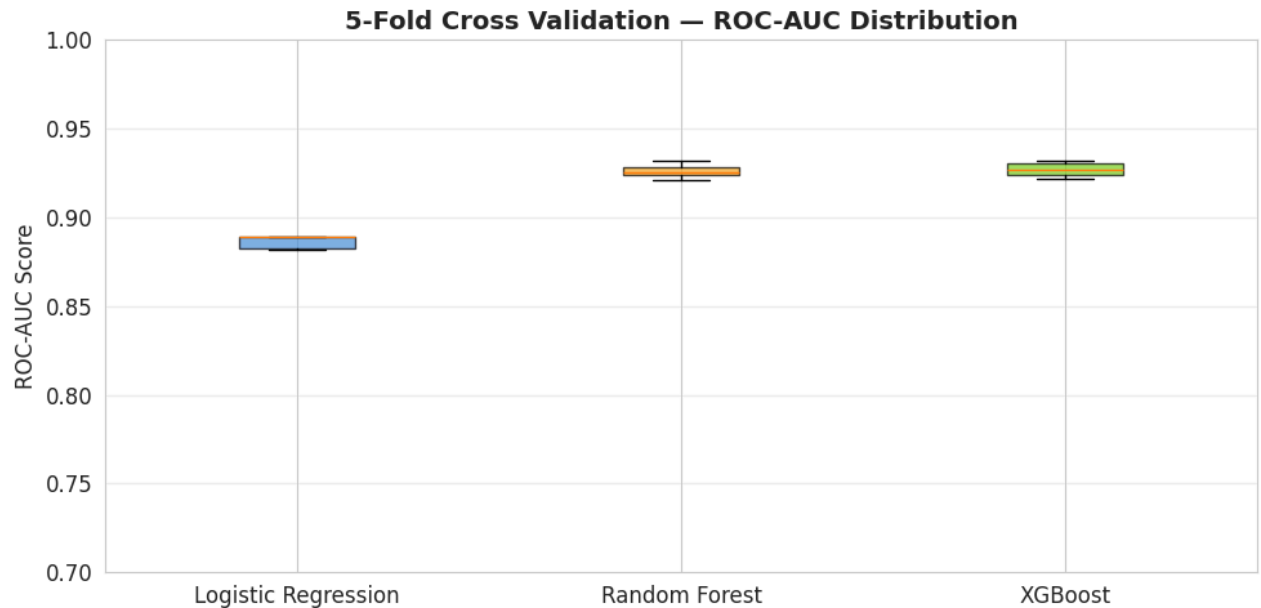
bp = plt.boxplot(cv_data, labels=labels, patch_artist=True)
for patch, color in zip(bp['boxes'], colors_cv):
    patch.set_facecolor(color)
    patch.set_alpha(0.7)

plt.title('5-Fold Cross Validation — ROC-AUC Distribution',
         fontsize=14, fontweight='bold')
plt.ylabel('ROC-AUC Score')
plt.ylim(0.7, 1.0)
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('chart8_cross_validation.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ Cross Validation done!")
```



5-FOLD CROSS VALIDATION RESULTS (ROC-AUC)

=====
 CommunicationPlatform : 0.8861 ± 0.0034
 Random Forest : 0.9261 ± 0.0036
 XGBoost : 0.9272 ± 0.0037
 DedicatedSupport : 0.9272 ± 0.0037
 =====



✓ Cross Validation done!

Model Results & Interpretation

Cross Validation Results (5-Fold ROC-AUC)

- Logistic Regression : 0.8861 ± 0.0034
- Random Forest : 0.9261 ± 0.0036
- XGBoost : 0.9272 ± 0.0037 ★ Best

Key Findings

- **XGBoost** achieved the highest cross-validation ROC-AUC of 0.9272, meaning it correctly ranks a churning donor above a retained donor 92.7% of the time
- The low standard deviation (± 0.0037) confirms the model is **stable and not overfitting**
- All three models significantly outperform random guessing (AUC = 0.50)

Why ROC-AUC Matters Here

With 26.5% churn rate, accuracy alone is misleading. A model that predicts "retained" for every donor would score 73.5% accuracy but catch zero churners. ROC-AUC penalizes this behavior, making it the correct metric for this problem.

```
# SHAP EXPLAINABILITY
!pip install shap -q

import shap

# Use XGBoost for SHAP
explainer = shap.TreeExplainer(xgb)
shap_values = explainer.shap_values(X_test)

# --- Chart 9: SHAP Summary Plot ---
plt.figure()
shap.summary_plot(shap_values, X_test,
                  feature_names=X.columns.tolist(),
                  show=False)
plt.title('SHAP - Feature Impact on Donor Churn',
          fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('chart9_shap_summary.png', dpi=150, bbox_inches='tight')
plt.show()
print("✓ SHAP Summary Plot saved!")

# --- Chart 10: SHAP Bar Plot
plt.figure()
shap.summary_plot(shap_values, X_test,
```

```
        feature_names=X.columns.tolist(),
        plot_type='bar',
        show=False)
plt.title('SHAP - Mean Feature Importance',
          fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('chart10_shap_bar.png', dpi=150, bbox_inches='tight')
plt.show()
print("✅ SHAP Bar Plot saved!")
```


SHAP — Feature Impact on Donor Churn



```
# DONOR RISK SCORING

# Get churn probability for every donor in test set
churn_probs = xgb.predict_proba(X_test)[: , 1]

# Create risk score dataframe
risk_df = pd.DataFrame({
    'ChurnProbability': churn_probs,
    'RiskScore': (churn_probs * 100).round(1),
    'ActualChurn': y_test.values
})

# Assign risk category
def assign_risk(prob):
    if prob >= 0.70:
        return '● High Risk'
    elif prob >= 0.40:
        return '● Medium Risk'
    else:
        return '● Low Risk'

risk_df['RiskCategory'] = risk_df['ChurnProbability'].apply(assign_risk)

# Summary
print("\n DONOR RISK SCORE DISTRIBUTION")
print("="*45)
print(risk_df['RiskCategory'].value_counts().to_string())
print("="*45)
print(f"\nAverage churn probability: {churn_probs.mean()*100:.1f}%")
print(f"Highest risk donor score : {churn_probs.max()*100:.1f}%")
print(f"Lowest risk donor score : {churn_probs.min()*100:.1f}%")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

risk_counts = risk_df['RiskCategory'].value_counts()
colors_risk = ['#E05C5C', '#F5A623', '#7ED321']
axes[0].bar(risk_counts.index, risk_counts.values, color=colors_risk)
```